INFO 5602
Information Visualization

Why U.S. flights are always late
(Dashboard Report)

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Context

- Invention of the plane until the late 1970s, air travel was regulated
  - Prices were managed by the U.S. government
  - Airlines couldn’t compete
  - “Had” to care about passengers

- Deregulation of U.S. airlines in 1978
  - Opened airlines to compete in the free market
  - Made travel more accessible to the average person
  - Made ticket prices cheaper
  - Cut all the good commodities
  - Cut functional stuff as well

Motivation

- Modern air travel is used by everyone for various reasons
  - Vacations
  - Business
  - Family
- Increased vocal testimonies (and personal experience) of poor air travel in the past couple of years than ever before
  - Annoying add-ons
  - Shrinking carry-on space
  - Increasing ticket prices
- Delays impact everyone (regardless of how rich you are)
  - A person in first-class is still waiting the same amount of time as a person in economy

Research questions

1. What is the biggest cause of flight delays?
   - How many delayed flights? How many minutes does each delay cost?

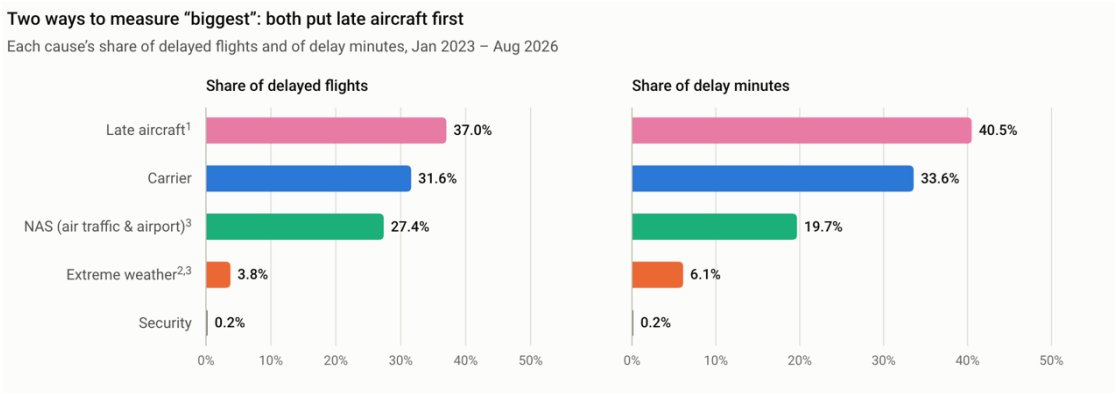
2. Does a busier time of year mean more delays?
   - Holidays? Vacation?

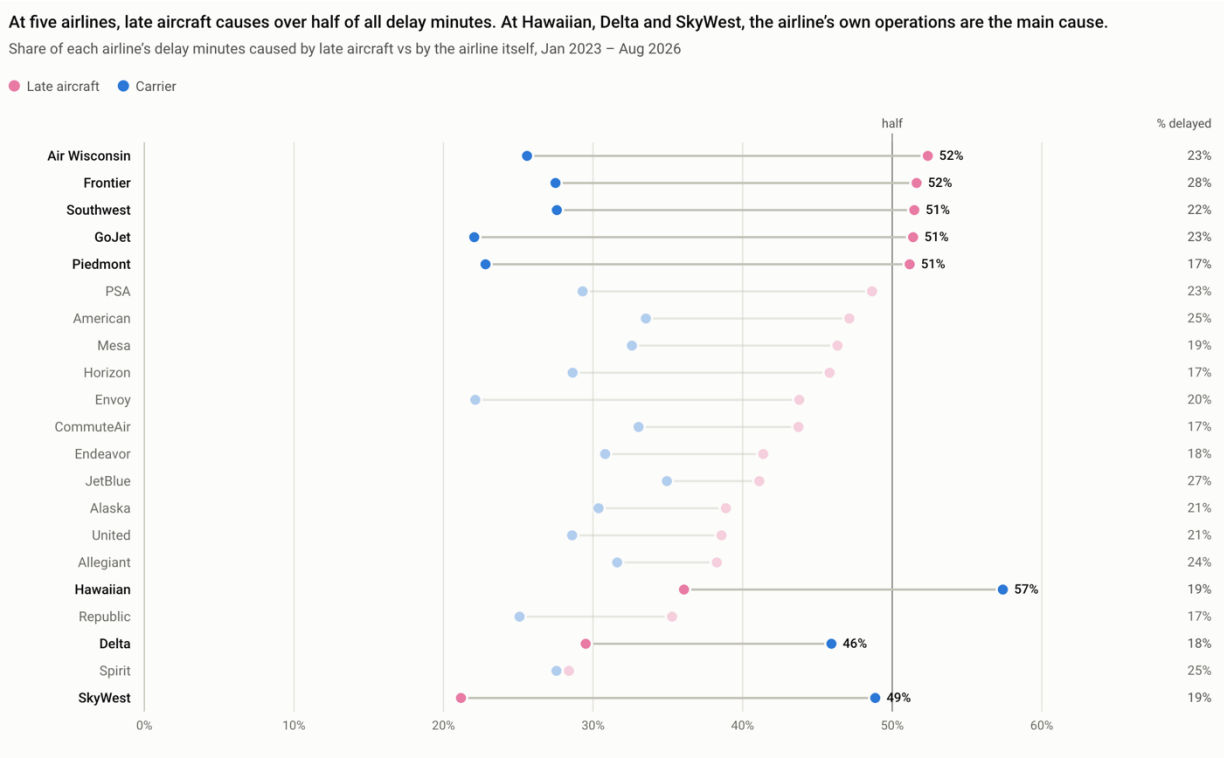
3. Which airports experience the most delays?
   - Without invoking: https://xkcd.com/1138/

Data source

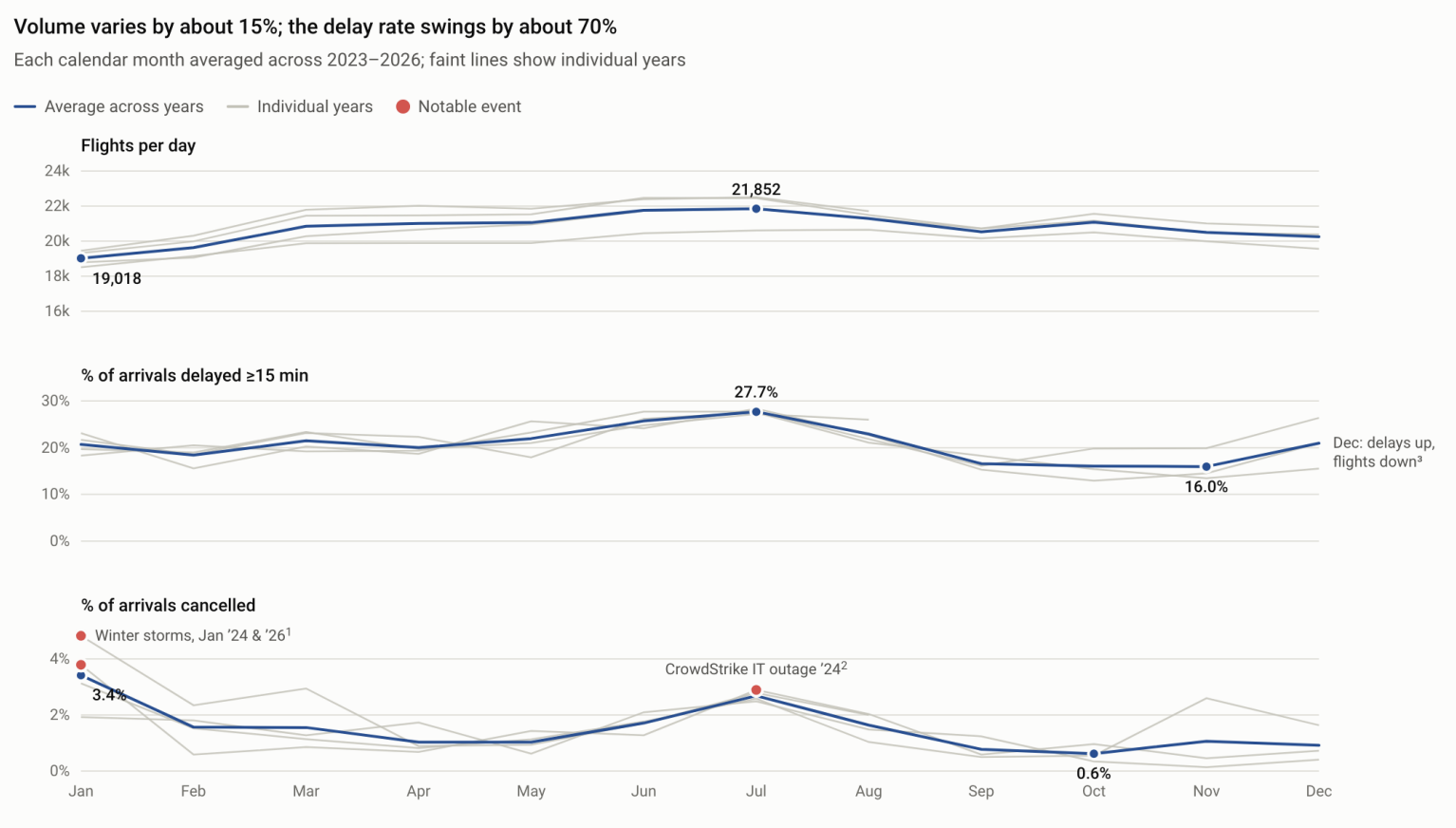
- **Source:**
  - Bureau of Transportation Statistics Airline On-Time Statistics and Delay Causes
  - Aggregation of all U.S. flights since June 2003
  - Looking at specifically after COVID flights from 2023 to 2026
- **Limitations:**
  - This dataset aggregates flights monthly, so no time, or per day data for each flight
  - Only for domestic U.S. flights; global patterns cannot be observed
  - NAS definitions of “extreme-weather” does not encapsulate all weather-related issues
  - Data is only for arriving flights to a specific airport.
    - A delay at SFO might have originated elsewhere.
- **Prep:**
  - Each row was one flight
  - Split all data into separate sheets: carrier, airports, causes, month

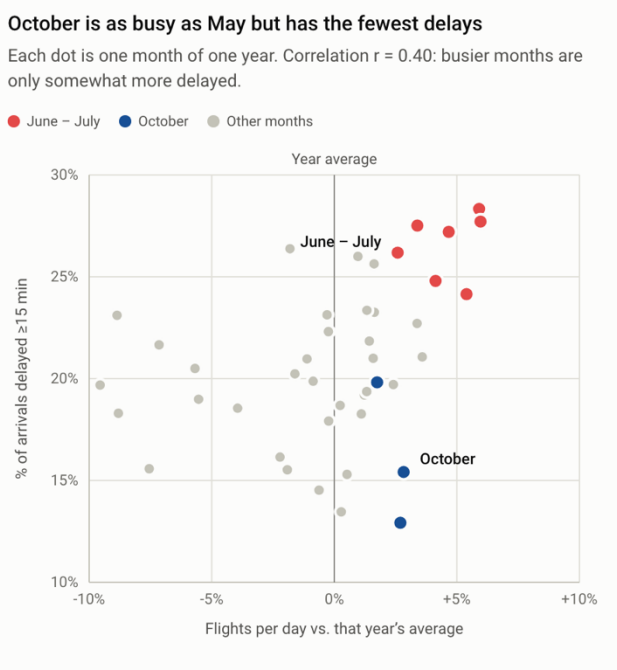
Charts – Q1

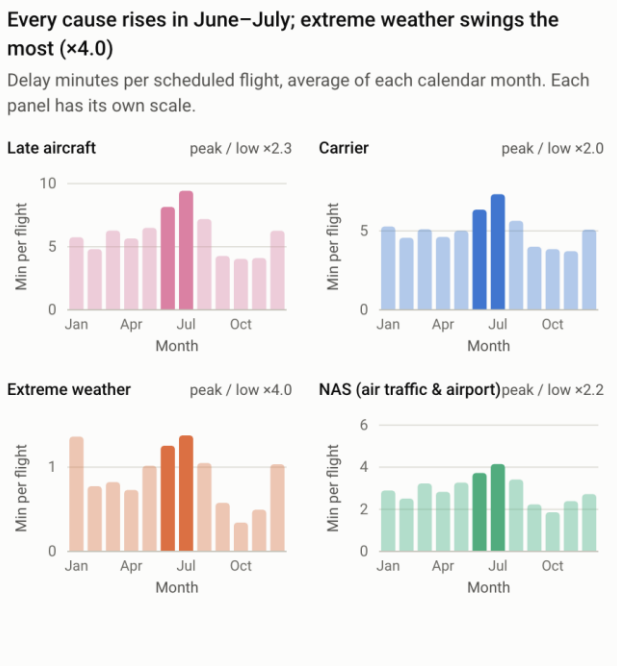

Two ways to measure “biggest”: both put late aircraft first
Each cause’s share of delayed flights and of delay minutes, Jan 2023 – Aug 2026
Share of delayed flights
Share of delay minutes
Late aircraft[1]
Carrier
NAS (air traffic & airport)[3]
Extreme weather[2,3]
Security
37.0%
31.6%
27.4%
3.8%
0.2%
40.5%
33.6%
19.7%
6.1%
0.2%
0%
10%
20%
30%
40%
50%


At five airlines, late aircraft causes over half of all delay minutes. At Hawaiian, Delta and SkyWest, the airline’s own operations are the main cause.
Share of each airline’s delay minutes caused by late aircraft vs by the airline itself, Jan 2023 – Aug 2026
Late aircraft
Carrier
half
% delayed
Air Wisconsin 52% 23%
Frontier 52% 28%
Southwest 51% 22%
GoJet 51% 23%
Piedmont 51% 17%
PSA 23%
American 25%
Mesa 19%
Horizon 17%
Envoy 20%
CommuteAir 17%
Endeavor 18%
JetBlue 27%
Alaska 21%
United 21%
Allegiant 24%
Hawaiian 57% 19%
Republic 17%
Delta 46% 18%
Spirit 25%
SkyWest 49% 19%
0%
10%
20%
30%
40%
50%
60%

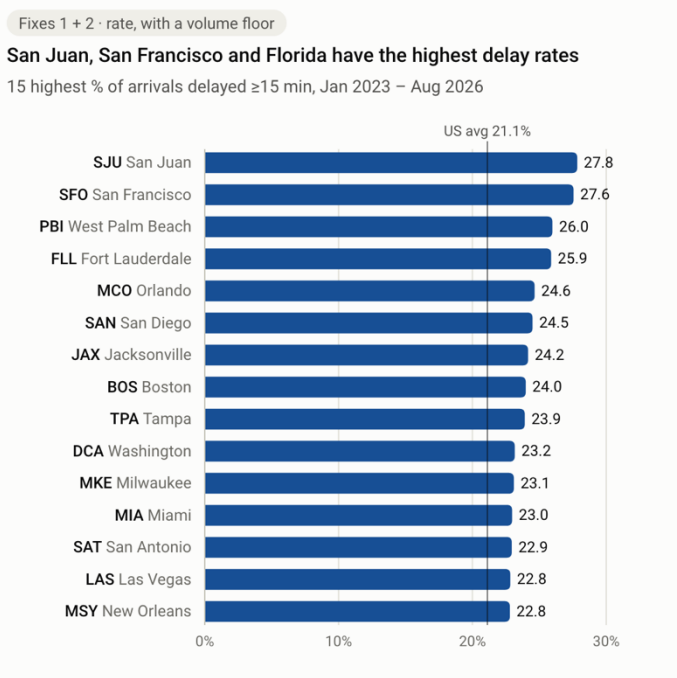
Charts – Q2

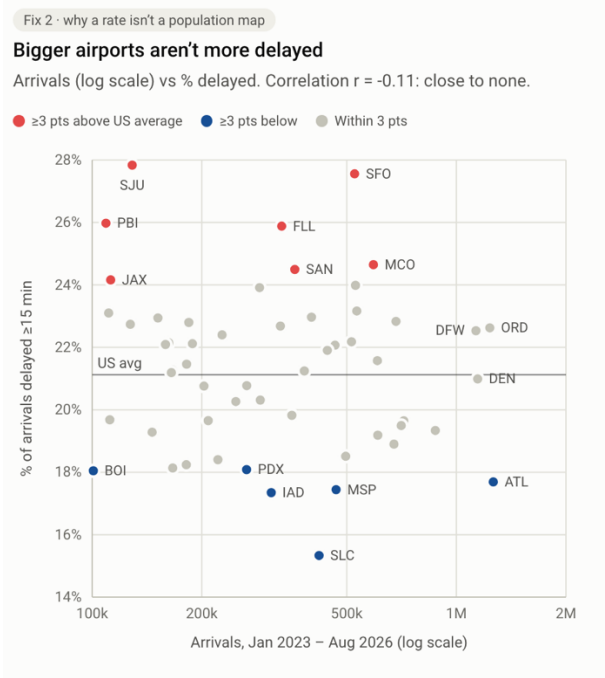

Volume varies by about 15%; the delay rate swings by about 70%
Each calendar month averaged across 2023–2026; faint lines show individual years
Average across years
Individual years
Notable event
Flights per day
24k
22k
20k
18k
16k
21,852
19,018
% of arrivals delayed ≥15 min
30%
20%
10%
0%
27.7%
16.0%
Dec: delays up, flights down[3]
% of arrivals cancelled
Winter storms, Jan '24 & '26[1]
4%
2%
0%
3.4%
CrowdStrike IT outage '24[2]
0.6%
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec

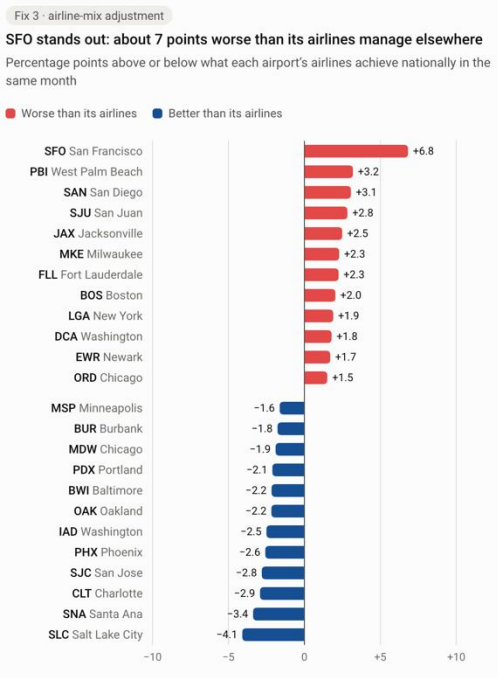

October is as busy as May but has the fewest delays
Each dot is one month of one year. Correlation r = 0.40: busier months are only somewhat more delayed.
June – July
October
Other months
Year average
30%
25%
20%
15%
10%
% of arrivals delayed ≥15 min
June – July
October
-10%
-5%
0%
+5%
+10%
Flights per day vs. that year's average

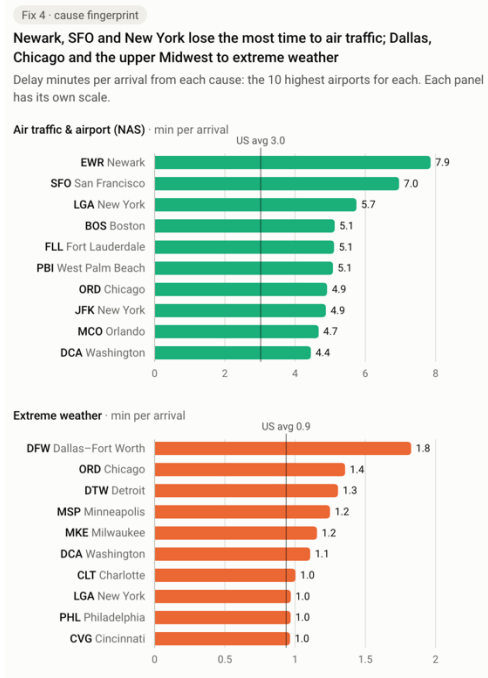

Every cause rises in June–July; extreme weather swings the most (×4.0)
Delay minutes per scheduled flight, average of each calendar month. Each panel has its own scale.
Late aircraft
peak / low ×2.3
Carrier
peak / low ×2.0
Extreme weather
peak / low ×4.0
NAS (air traffic & airport)
peak / low ×2.2
Min per flight
Jan
Apr
Jul
Oct
Month

Charts – Q3


Fixes 1 + 2 · rate, with a volume floor
San Juan, San Francisco and Florida have the highest delay rates
15 highest % of arrivals delayed ≥15 min, Jan 2023 – Aug 2026
US avg 21.1%
SJU San Juan 27.8
SFO San Francisco 27.6
PBI West Palm Beach 26.0
FLL Fort Lauderdale 25.9
MCO Orlando 24.6
SAN San Diego 24.5
JAX Jacksonville 24.2
BOS Boston 24.0
TPA Tampa 23.9
DCA Washington 23.2
MKE Milwaukee 23.1
MIA Miami 23.0
SAT San Antonio 22.9
LAS Las Vegas 22.8
MSY New Orleans 22.8
0%
10%
20%
30%


Fix 2 · why a rate isn't a population map
Bigger airports aren't more delayed
Arrivals (log scale) vs % delayed. Correlation r = -0.11: close to none.
≥3 pts above US average
≥3 pts below
Within 3 pts
% of arrivals delayed ≥15 min
28%
26%
24%
22%
20%
18%
16%
14%
SJU
SFO
PBI
FLL
SAN
MCO
JAX
DFW
ORD
US avg
DEN
BOI
PDX
IAD
MSP
ATL
SLC
100k
200k
500k
1M
2M
Arrivals, Jan 2023 – Aug 2026 (log scale)


Fix 3 · airline-mix adjustment
SFO stands out: about 7 points worse than its airlines manage elsewhere
Percentage points above or below what each airport's airlines achieve nationally in the same month
Worse than its airlines
Better than its airlines
SFO San Francisco +6.8
PBI West Palm Beach +3.2
SAN San Diego +3.1
SJU San Juan +2.8
JAX Jacksonville +2.5
MKE Milwaukee +2.3
FLL Fort Lauderdale +2.3
BOS Boston +2.0
LGA New York +1.9
DCA Washington +1.8
EWR Newark +1.7
ORD Chicago +1.5
MSP Minneapolis −1.6
BUR Burbank −1.8
MDW Chicago −1.9
PDX Portland −2.1
BWI Baltimore −2.2
OAK Oakland −2.2
IAD Washington −2.5
PHX Phoenix −2.6
SJC San Jose −2.8
CLT Charlotte −2.9
SNA Santa Ana −3.4
SLC Salt Lake City −4.1
−10
−5
0
+5
+10


Fix 4 · cause fingerprint
Newark, SFO and New York lose the most time to air traffic; Dallas, Chicago and the upper Midwest to extreme weather
Delay minutes per arrival from each cause: the 10 highest airports for each. Each panel has its own scale.
Air traffic & airport (NAS) · min per arrival
US avg 3.0
EWR Newark 7.9
SFO San Francisco 7.0
LGA New York 5.7
BOS Boston 5.1
FLL Fort Lauderdale 5.1
PBI West Palm Beach 5.1
ORD Chicago 4.9
JFK New York 4.9
MCO Orlando 4.7
DCA Washington 4.4
0
2
4
6
8
Extreme weather · min per arrival
US avg 0.9
DFW Dallas–Fort Worth 1.8
ORD Chicago 1.4
DTW Detroit 1.3
MSP Minneapolis 1.2
MKE Milwaukee 1.2
DCA Washington 1.1
CLT Charlotte 1.0
LGA New York 1.0
PHL Philadelphia 1.0
CVG Cincinnati 1.0
0
0.5
1
1.5
2

Dashboard

This slide was left intentionally blank as an exercise to the reader.

Answers

1. What is the biggest cause of delays?
   - Late aircrafts, but that is more like a domino effect
   - The main source is the airline's own operations

2. Which time of year is the busiest, and does busy mean more delays?
   - Domestic flight volume barely changes throughout the year
   - Delays double in the summer. Cause? Extreme weather

3. Which airports experience the most delays?
   - Looking at airports with more than 100k arrivals, San Juan , SF, and West Palm Beach have the worst
     - Airlines have more success at other airports than these ones
   - Newark, San Francisco, New York, lost time due to air traffic
   - Dallas Fort-Worth, Chicago, Detroit, lost time due to extreme weather

URLS

- https://info-5602-dashboard.vercel.app/
- https://github.com/rfhcs/info-5602-dashboard

Sources

- Answer In Progress. Flying is getting worse, here's why. https://www.youtube.com/watch?v=Aa3vkKRCczg
- *OT delay*. Bureau of Transportation Statistics. https://www.transtats.bts.gov/ot_delay/OT_DelayCause1.asp?20=E
- Robert Peterson. Impacts of Airline Deregulation. https://onlinepubs.trb.org/onlinepubs/trnews/trnews315airlinedereg.pdf
- Maxwell Tabarrok. Is Air Travel Getting Worse?. https://www.maximum-progress.com/p/is-air-travel-getting-worse